Outcome Proof: SonicAtlas: An Interactive In-Browser Exploratory Data Analysis Platform for Multi-Dimensional Audio Features and Lyrics of Spotify Tracks Across Genres and Eras

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TABLE I
IEEE FORMAT COMPLIANCE CHECK (AUTOMATED, DOCS\{}_BUILD/IEEE\{}_CHECK.PY)

Check	Result
Two-column layout	yes (5 of 6 pages with two text columns)
Page size	A4, 595.3 x 841.9 pt (pass)
Length	6 pages (6-8 required)
Fonts embedded	all 5 fonts embedded
Type-3 fonts	none
Text margins (pt)	L 40.6, R 40.6, T 51.2
Page numbers / headers	none (as the template requires)
Figure resolution	minimum 566 dpi effective (>= 300)
References	30 listed; all 30 in-text citations resolved
IEEE PDF eXpress	pending - needs the conference ID; done by the team

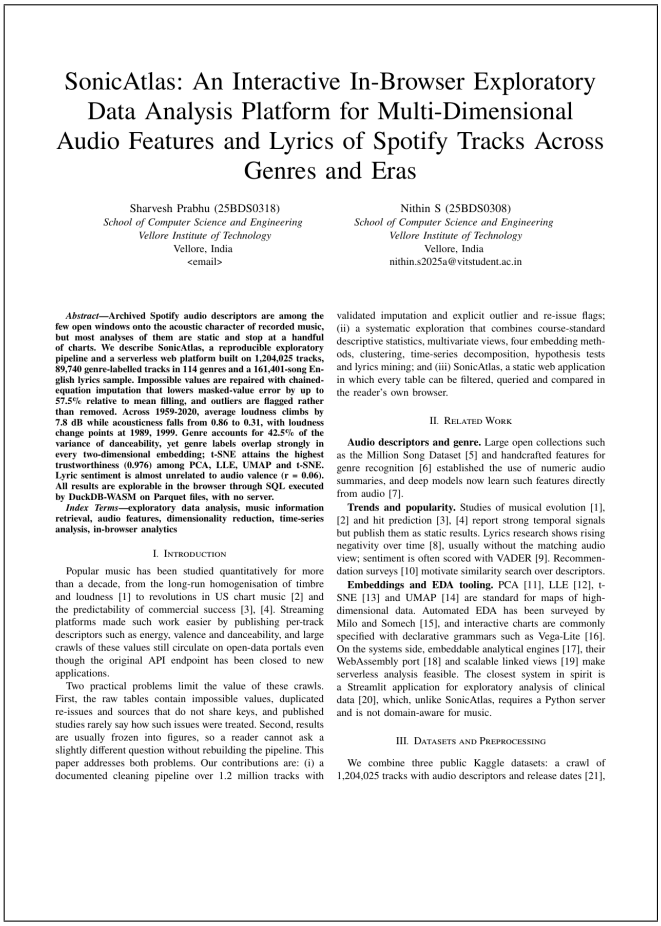
***Abstract*—This evidence pack documents the verifiable outcomes of the SonicAtlas course project: an IEEE-format research paper prepared for submission, an open-source code repository, a static website, reproducibility evidence, the key results, and an originality self-check. Every item can be checked against the files and links listed. Items that only the team can obtain later (submission receipt, acceptance, certificate, official Turnitin report) are left as empty, labelled slots.**

I. IEEE Research Paper

Fig. 1 shows the first page of the paper. Full citation (IEEE style):

> S. Prabhu and N. S, "SonicAtlas: An Interactive In-Browser Exploratory Data Analysis Platform for Multi-Dimensional Audio Features and Lyrics of Spotify Tracks Across Genres and Eras," Team 05 project paper, BACSE301 Exploratory Data Analysis, Vellore Institute of Technology, Vellore, India, Oct. 2026 (prepared for submission).

Status: prepared for submission. It has not been submitted, accepted or published. Target venues: IEEE CONECCT 2027 (https://www.ieee-conecct.org.ieeebangalore.org/); ISMIR (https://ismir2026.ismir.net/); IEEE VIS (https://ieeevis.org/year/2026/welcome/).



SonicAtlas: An Interactive In-Browser Exploratory Data Analysis Platform for Multi-Dimensional Audio Features and Lyrics of Spotify Tracks Across Genres and Eras

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***Abstract*—Archived Spotify audio descriptors are among the few open windows onto the acoustic character of recorded music, but most analyses of them are static and stop at a handful of charts. We describe SonicAtlas, a reproducible exploratory pipeline and a serverless web platform built on 1,204,025 tracks, 89,740 genre-labelled tracks in 114 genres and a 161,401-song English lyrics sample. Impossible values are repaired with chained-equation imputation that lowers masked-value error by up to 57.5% relative to mean filling, and outliers are flagged rather than removed. Across 1959-2020, average loudness climbs by 7.8 dB while acousticness falls from 0.86 to 0.31, with loudness change points at 1989, 1999. Genre accounts for 42.5% of the variance of danceability, yet genre labels overlap strongly in every two-dimensional embedding; t-SNE attains the highest trustworthiness (0.976) among PCA, LLE, UMAP and t-SNE. Lyric sentiment is almost unrelated to audio valence ($r \approx 0.06$). All results are explorable in the browser through SQL executed by DuckDB-WASM on Parquet files, with no server.**

***Index Terms*—exploratory data analysis, music information retrieval, audio features, dimensionality reduction, time-series analysis, in-browser analytics**

I. Introduction

Popular music has been studied quantitatively for more than a decade, from the long-run homogenisation of timbre and loudness [1] to revolutions in US chart music [2] and the predictability of commercial success [3], [4]. Streaming platforms made such work easier by publishing per-track descriptors such as energy, valence and danceability, and large crawls of these values still circulate on open-data portals even though the original API endpoint has been closed to new applications.

Two practical problems limit the value of these crawls. First, the raw tables contain impossible values, duplicated re-issues and sources that do not share keys, and published studies rarely say how such issues were treated. Second, results are usually frozen into figures, so a reader cannot ask a slightly different question without rebuilding the pipeline. This paper addresses both problems. Our contributions are: (i) a documented cleaning pipeline over 1.2 million tracks with validated imputation and explicit outlier and re-issue flags; (ii) a systematic exploration that combines course-standard descriptive statistics, multivariate views, four embedding methods, clustering, time-series decomposition, hypothesis tests and lyrics mining; and (iii) SonicAtlas, a static web application in which every table can be filtered, queried and compared in the reader's own browser.

II. Related Work

Audio descriptors and genre. Large open collections such as the Million Song Dataset [5] and handcrafted features for genre recognition [6] established the use of numeric audio summaries, and deep models now learn such features directly from audio [7].

Trends and popularity. Studies of musical evolution [1], [2] and hit prediction [3], [4] report strong temporal signals but publish them as static results. Lyrics research shows rising negativity over time [8], usually without the matching audio view; sentiment is often scored with VADER [9]. Recommendation surveys [10] motivate similarity search over descriptors.

Embeddings and EDA tooling. PCA [11], LLE [12], t-SNE [13] and UMAP [14] are standard for maps of high-dimensional data. Automated EDA has been surveyed by Milo and Somech [15], and interactive charts are commonly specified with declarative grammars such as Vega-Lite [16]. On the systems side, embeddable analytical engines [17], their WebAssembly port [18] and scalable linked views [19] make serverless analysis feasible. The closest system in spirit is a Streamlit application for exploratory analysis of clinical data [20], which, unlike SonicAtlas, requires a Python server and is not domain-aware for music.

III. Datasets and Preprocessing

We combine three public Kaggle datasets: a crawl of 1,204,025 tracks with audio descriptors and release dates [21],

Fig. 1. First page of the IEEE-format paper (IEEEtran conference class, A4).

II. Software Outcome

Repository: [GitHub Link]. Live website: [Live Website Link]. Licence: MIT (code). Local git history: 2 commits from 2026-10-20 to 2026-10-20; release tag: v1.0.0. The site is a static build (web/dist) deployed by the GitHub Actions workflow .github/workflows/deploy-pages.yml. Figures 2-6 are

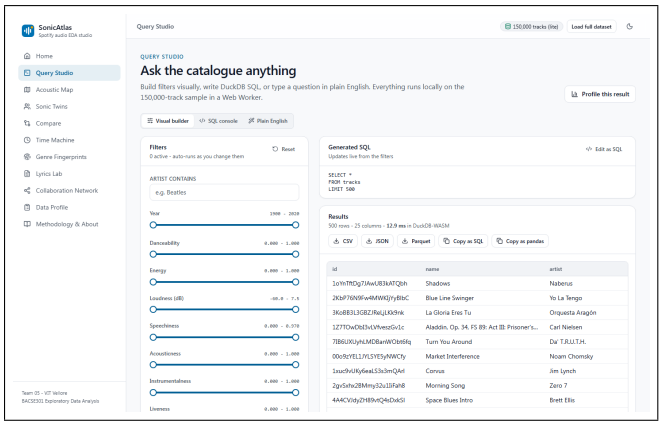


Fig. 2. Query Studio: SQL console and filter builder running on DuckDB-WASM.

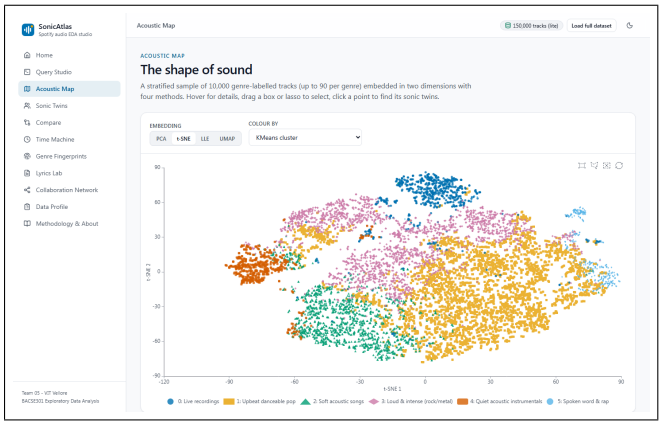


Fig. 3. Acoustic Map: switchable PCA / t-SNE / LLE / UMAP projections.

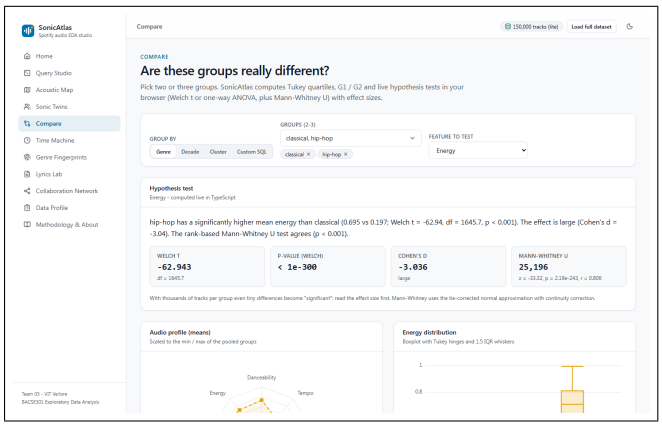


Fig. 4. Compare: group statistics with live significance tests.

real screenshots of the running site.

III. Reproducibility

TABLE II lists the evidence that the project can be rebuilt from scratch.

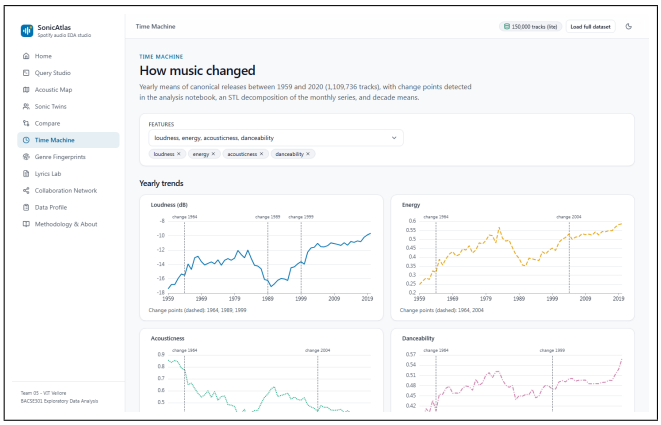


Fig. 5. Time Machine: yearly trends, STL components and change points.

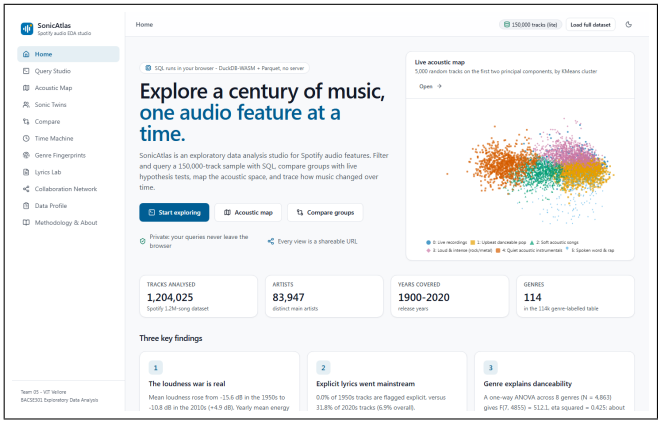


Fig. 6. Home page with headline statistics and key findings.

TABLE II
Reproducibility evidence

Item	Evidence
Python environment	Python 3.13, pinned requirements.txt (+ requirements-profiling.txt)
Notebooks	10 notebooks run top to bottom (notebooks/run_log.txt)
Python unit tests	7/7
Web unit tests	65/65
End-to-end tests	8/8
Lighthouse	performance 98-99, accessibility 100, best practices 100, SEO 100 on four pages
Lite-data query time	4.7 ms median
Number cross-check	all key numbers in report, paper and decks equal results.json (docs_build/check_numbers.py)

IV. Results Summary

TABLE III repeats the key numbers from reports/results.json; Figs. 7-9 show three key figures.

V. Originality

A similarity self-check (not an official Turnitin report) compared every document with a corpus of 57 sources (93,391 words) by 8-word fingerprints and TF-IDF sentence cosine,

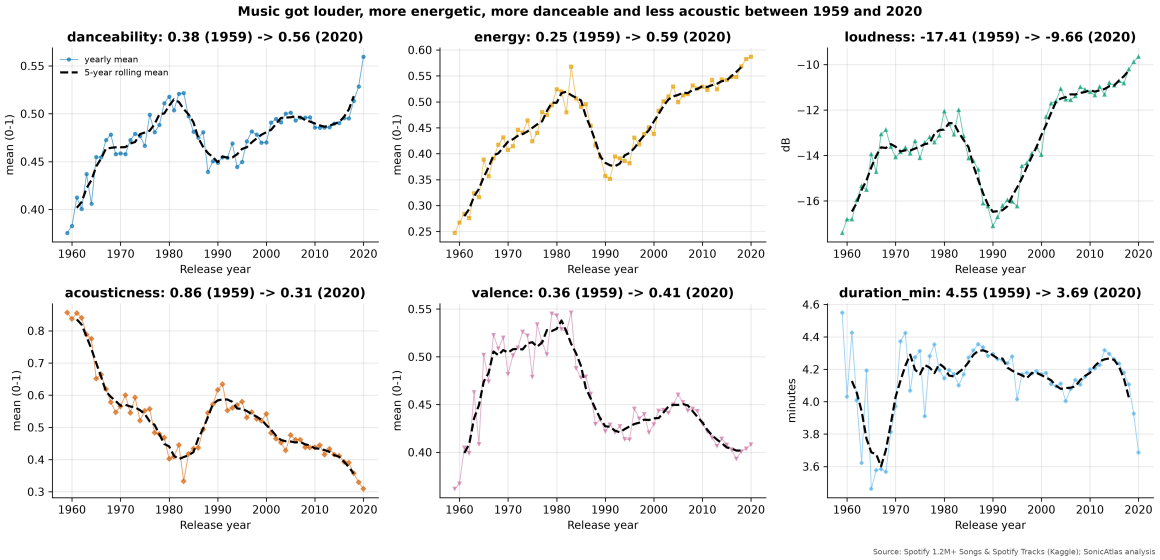


Fig. 7. Yearly means and 5-year rolling means of six audio descriptors.

TABLE III
KEY RESULTS (FROM RESULTS.JSON)

Quantity	Value
Tracks analysed	1,204,025
Genre-labelled tracks / genres	89,740 / 114
Zero-BPM tempos repaired (MICE)	2,777
Loudness change	-17.4 to -9.7 dB (1959-2020)
Loudness ~ energy (Pearson)	0.82
Genre effect on danceability (eta sq.)	0.425
t-SNE trustworthiness	0.976
Lyric sentiment ~ valence (r)	0.06

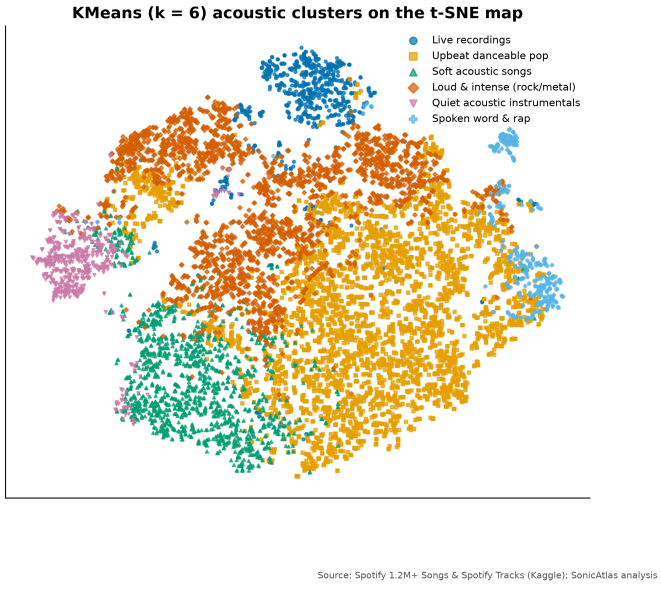


Fig. 8. Six named KMeans clusters on the t-SNE map.

TABLE IV
SIMILARITY SELF-CHECK RESULTS

Document	Similarity	Max one source
Project Report	0.18%	0.09%
IEEE Paper	0.00%	0.00%
Review2 PPT	0.00%	0.00%
Review3 PPT	0.00%	0.00%
Final PPT	0.00%	0.00%
Review1 PPT	0.00%	0.00%

and searched 90 sampled sentences on the web as exact phrases (prose matches: report 1 - a cited paper title - and paper 0). Details are in the Plagiarism Report.

VI. PENDING PROOFS (TO BE ADDED BY THE TEAM)

The following evidence does not exist yet and must NOT be created by anyone other than the issuing body. Each slot is intentionally empty.

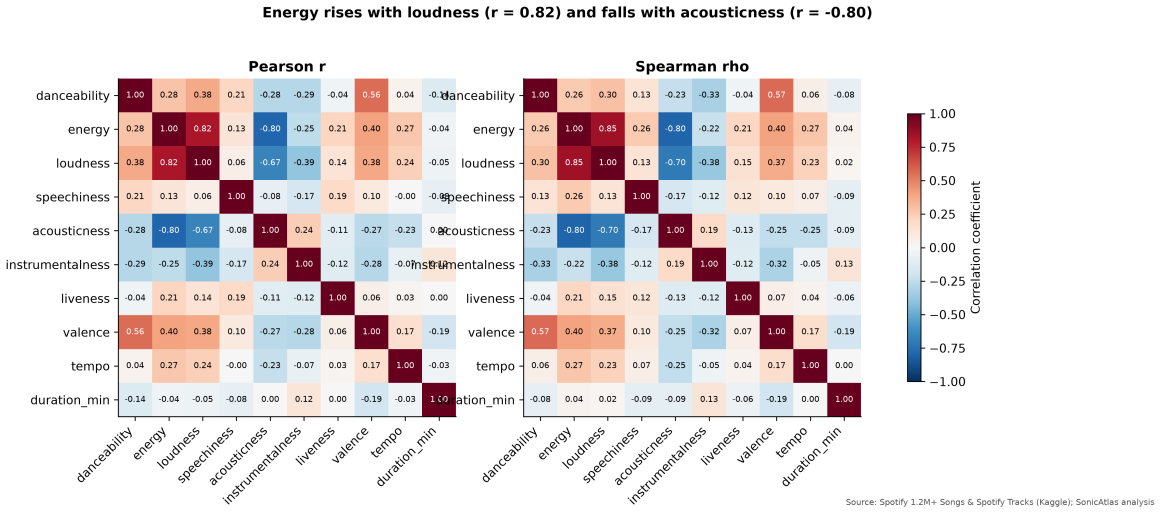


Fig. 9. Pearson and Spearman correlation matrices.

TABLE V
PENDING PROOFS

Conference submission acknowledgement / paper ID	
Acceptance e-mail	
Presentation certificate	
Official Turnitin report from VIT	
IEEE PDF eXpress compliance report	